

Ostin Christopher Torres Galeas

Josselyn Andrea Ulcuango Quimbiamba

Mateo Jhaer Unda Piguave

Yandry Josue Velez Valencia

University of the Armed Forces - ESPE

NRC: 36862. Object Oriented Programming

PhD Lascano Jorge Edison

Project Name: LegalSync
Team Name: CodeLex

1. **User:** A base entity representing any member of the law firm, with encapsulated attributes (userId, name, email, password, role, active).
2. **Lawyer:** A subclass of User representing the firm's lawyers (Director, Partner, Associate, and Junior), including their license number, position, and operations for issuing directives or managing access rights.
3. **LegalAssistant:** A subclass of User representing a legal assistant linked to a judicial unit (judicialUnit), responsible for updating assigned case proceedings.
4. **Secretary:** A subclass of User with permissions restricted to viewing case status, calendar schedules, and expense/billing reports.
5. **Client:** An entity that stores identification and contact information for a client represented by the firm (clientId, idNumber, name, phone, email, address).
6. **Defendant:** An entity that stores information regarding the opposing party or defendant in a lawsuit (defendantId, idNumber, name, contactInfo, address).
7. **CaseFile:** A central composite object representing a judicial case file (caseId, docketNumber, title, processClass, category, priority, judicialUnit, status, openingDate, client, defendant, assignedLawyer, and lists of documents, evidence, deadlines, and expenses).
8. **Expense:** An object representing a procedural cost or payment item associated with a case (expenseId, caseId, description, amount, expenseDate, paid), used to calculate the client's outstanding balance.
9. **Deadline:** An object representing a judicial date, hearing, directive from a lawyer, or pending task (deadlineId, caseId, description, disposition, type, targetDate, assignedUser, completed).

10. **Document and Evidence:** Objects that store metadata and the local file path (filePath) for legal documents and evidence linked to a case file.
11. **AuditEntry and AuditLog:** Objects responsible for recording and storing every action or modification performed by a user, including the exact date and time. Enumerations (Role, LegalCategory, CaseStatus, Priority): Data types used to restrict the 6 roles, legal categories, case status (Active/Passive/In Progress/Closed), and priority.
12. **Service and Persistence Objects (AuthenticationService, SearchEngine, JsonFileManager, CsvReportGenerator, ConsoleMenu):** Classes responsible for authentication logic, multi-criteria filtering within collections, reading/writing .json files, generating .csv reports, and console-based keyboard navigation.